

Vaibav Malpeddi

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EDUCATION

Rice University

Master of Computer Science - (GPA: 3.90/4.00)

Relevant Coursework: Software Engineering Methodology, Design Analysis of Algorithms, Deep Learning

Houston, TX

Dec 2025

SRM University

Bachelor of Technology in Computer Science - (GPA: 3.90/4.00)

India

Sep 2020 - Jun 2024

SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, Go

Web & APIs: React, Spring Boot, Node.js, Flask, FastAPI, REST APIs, GraphQL, API Integration, HTML/CSS

Databases: SQL, PostgreSQL, Bigtable, Redis

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD

Data & ML: TensorFlow, scikit-learn, pandas, NumPy, FAISS, CUDA

Tools & Practices: Git, Bash, debugging, troubleshooting, unit/integration testing, code reviews, Agile/Scrum

WORK EXPERIENCE

Software Engineer Intern – AI/Machine Learning

Affekta

Houston, TX

Oct 2025 - Dec 2025

- Built and validated an evaluation framework for detecting **“Unknown Unknowns” in space-habitat telemetry**, assessing **6+** anomaly-detection models across **10+** datasets and **40+** injected fault scenarios for **NASA STTR Phase II research**
- Translated turbofan engine anomaly datasets into **deep-space habitat analogs** by mapping **25+** sensor features and fault signatures across domains, enabling cross-dataset generalization tests and reducing domain-specific false alerts by **30%**
- Designed and verified **synthetic fault scenarios (drift, bias, stuck-at, and composite failures)** generating **150+** labeled evaluation cases to stress-test model robustness under previously unseen failure modes

Software Engineer Intern – AI/Robotics

Rice University

Houston, TX

Jun 2025 - Aug 2025

- Delivered a high-fidelity **digital twin system for 90+ indoor scenes** by integrating 10,800 RGB-D frames using ROS Hydra and Genesis, leveraging CUDA-accelerated kernels to achieve **under 5 cm reconstruction accuracy**
- Reduced end-to-end development cycles by **30%** through roadmap design, automation, and streamlined I/O workflows
- Developed **modular, production-style code with unit tests and end-to-end tests** to ensure stability during rapid deployments

Natural Language Processing Intern

The University of Winnipeg

Winnipeg, Canada

Mar 2023 - Aug 2023

- Improved non-topic classification accuracy by **12%** on **50,000+** records by integrating XLNet into existing NLP workflows
- Built scalable **client-server data pipelines**, reducing system latency by **40%**, and supported repeated model retraining
- Maintained **CI pipelines, debugging workflows, Git version control, and code review practices**, improving system reliability

ACADEMIC PROJECTS

URLino - Advanced URL Shortener Platform

Feb 2025 - Apr 2025

- Built a distributed high-availability **full-stack web application** through **React, Spring Boot, and Bigtable** that delivered **600-800 ms shortened latency, under 200 ms redirects, and 99.99% uptime**
- Led a **6-person team** implementing **JWT-secured REST APIs, analytics dashboards, reverse lookup services, and CI/CD pipelines**, maintaining a failure rate **under 1%** while enabling rapid releases

TicketTron - Online Movie Ticket Booking Platform

Feb 2025 - Apr 2025

- Built a **full-stack booking platform** using **React, HTML, CSS, JavaScript, Spring Boot, and SQL**, enabling users to browse movies, select seats, and complete bookings through a responsive interface, tested with **1,000+ simulated user sessions**
- Designed secure transaction workflows with **validation, error handling, and concurrency-safe backend services**, achieving **>99% successful transaction completion** under **100+ parallel requests**

RAGBot - Retrieval-Augmented Chatbot

Feb 2025 - Apr 2025

- Created a **retrieval pipeline over 1M+ Reddit posts** through FAISS and MiniLM vector indexing with query re-ranking and caching that improved **top-5 relevance by 25%** while sustaining latency under **300 ms**
- Deployed a scalable **client-server AI system** with DeepSeek Qwen-1.5B through continuous deployment pipelines and automated rollout checks that boosted user engagement by **40%**